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and explainable artificial intelligence (XAI) methods to provide an interpretable machine learning framework for prediction road traffic accidents. The proposed methodology framework uses statistical analyze and pre-processing techniques (missing value treatment, normalization, feature encoding, and class balancing) to improve data quality and reduce bias. The final preprocessed data set contained 1180 observations, 14 numerical predictor variables and a binary response variable on the occurrence of the accident. To avoid overfit and to ensure a robust performance evaluation, ten supervised machine learning classifiers were built and evaluated using stratified 10-fold cross-validation. Accuracy, precision, recall and f1-score were uses to evaluate the model's performance. Experimental results prove that ensemble-based gradient boosting models outperformed the traditional baseline classifiers. Among all models, XGBoost achieved the highest prediction accuracy (94.3%), precision (93.8%), recall (94.1%), and F1-score (93.9%) and was a strong generalizer. The second best accuracy was LightGBM with (92.7%) and lower computing complexity. To improvement the interpretability of the model, we used SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) to provide both global and local explanations for the model predictions. The explainability research shows that various important factors affect the probability of accidents such as traffic density, vehicle speed, road features, and environmental conditions. The proposed framework can help traffic authorities and legislators in designing data-driven plans to improve road safety and provide a clear and reliable answer for intelligent transportation systems.

**Keywords:** Road traffic accidents, Machine Learning, Ensemble learning, XGBoost, LightGBM, Explainable artificial intelligence (XAI)

#### **Paper ID: 107**

### **A Multi-Scale Attention-Based Dense Residual Network for Sugarcane Leaf Disease Detection Using BSRI Data**

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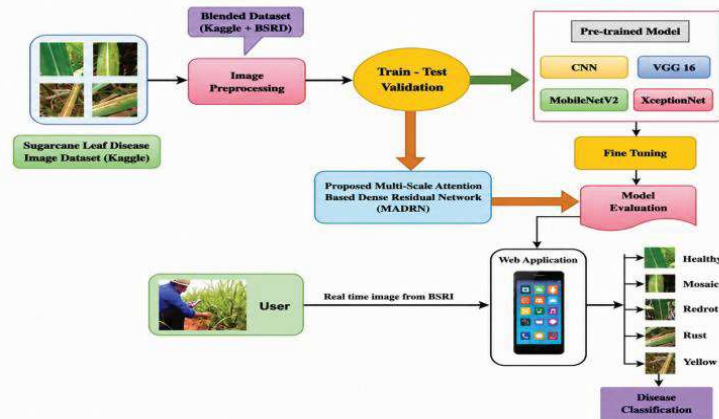
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#### **Abstract:**

Sugarcane is an economically important crop in Bangladesh, yet leaf diseases such as yellow leaf, rust, red rot, and mosaic can substantially reduce yield and quality. Accurate image-based disease diagnosis remains challenging because of high visual similarity among disease classes, real-world variability in leaf appearance, and limited annotated local data. To address these challenges, this study proposes an advanced deep learning framework for automated sugarcane leaf disease classification using a novel architecture, Multi-Scale Attention-Based Dense Residual Network (MADRN). The proposed architecture integrates dense residual learning with multi-scale attention to capture both fine-grained and global visual patterns, thereby improving discrimination among visually similar disease categories. The model was evaluated on two datasets: a publicly available Kaggle dataset comprising 2,521 images distributed across five classes (Healthy: 522, Mosaic: 462, Red Rot: 518, Rust: 514, and Yellow: 505) and a blended dataset created by combining Kaggle images with samples collected from the Bangladesh Sugarcrop Research Institute (BSRI), representing realistic field conditions. The datasets were partitioned into training, validation, and testing sets using a 70:15:15 ratio. The BSRI images were collected under diverse environmental conditions, including variations in illumination, humidity, plant growth stages, viewing angles, and background complexity, to better reflect real-world agricultural scenarios. Prior to training, all images were preprocessed through resizing, normalization, and data augmentation to improve robustness and generalization. For comparative assessment, several conventional and fine-tuned deep learning models, including CNN, VGG16, MobileNetV2, and XceptionNet, were evaluated alongside benchmarking against recent state-of-the-art sugarcane disease classification studies. Experimental results demonstrate that MADRN consistently outperforms competing models, achieving 94.78% accuracy and 99.00% AUC on the Kaggle dataset, and 92.25%

accuracy and 95.00% AUC on the blended dataset, together with strong precision, recall, and F1-score values. These findings indicate that the proposed architecture provides robust and effective feature learning for complex sugarcane leaf disease patterns and maintains strong generalization under diverse environmental conditions. To support practical deployment, a web-based application was developed for real-time, user-friendly disease detection. The proposed framework offers a reliable and scalable solution for early disease diagnosis and decision support in sugarcane cultivation.



**Keywords:** Sugarcane; Leaf Disease; Deep Learning; Attention Mechanism; MADRN.

**Paper ID: 108**

### **Automated Five-Stage Diabetic Retinopathy Grading Using a Leak-Safe and Explainable Deep-Handcrafted Fusion Framework**

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#### **Abstract:**

Diabetic retinopathy (DR) is a major cause of preventable vision loss, yet fine-grained five-stage grading (Grades 0-4) from retinal fundus photographs remains expert-dependent and difficult to scale in resource-limited screening settings. This study proposes an explainable deep-handcrafted fusion framework for automated five-stage DR grading and introduces a unified leak-safe cross-source benchmark by harmonizing the APTOS 2019 and IDRiD datasets to support evaluation under more realistic mixed-source screening variability. To minimize data leakage, within-subset and cross-dataset duplicate screening was performed while preserving the official IDRiD test split. The resulting benchmark contains 4,045 unique images, including 3,534 from APTOS, 408 from IDRiD Train, and 103 from IDRiD Test. The proposed five-stage, validation-driven framework enables controlled comparison across frozen deep features, handcrafted descriptors, feature fusion, fine-tuning, and fine-tuned fusion under a consistent evaluation protocol. Performance was assessed on fixed Train/Validation/Test splits using Accuracy, Macro-F1, one-vs-rest AUC, and Cohen's kappa, with uncertainty estimated from 1,000 bootstrap resamples. DenseNet121 and SIFT-BoVW emerged as the strongest deep and handcrafted components, respectively. The final model, combining fine-tuned DenseNet121 embeddings, SIFT-BoVW, and XGBoost, achieved held-out test accuracy of  $78.71\% \pm 1.65\%$ , Macro-F1 of  $64.13\% \pm 2.55\%$ , Cohen's kappa of  $67.88\% \pm 2.30\%$ , and OvR AUC of  $94.80\% \pm 0.59\%$ . Compared with three lightweight custom CNN baselines trained end-to-end on the same inputs, the proposed fusion model delivered a stronger overall balance of accuracy, agreement, and class